

Individual Activity 4: Design Your Own System Using UML

Part 1: System Idea & Recall

1. UML Diagrams and Descriptions

Use Case Diagram

A Use Case Diagram shows how users (actors) interact with a system. It identifies the main functions that each user can perform in the system.

Activity Diagram

An Activity Diagram shows the step-by-step flow of activities in a system. It explains how a process starts, what actions are done, and how it ends.

Class Diagram

A Class Diagram shows the structure of a system. It includes classes, attributes, methods, and how each class is related to one another.

Sequence Diagram

A Sequence Diagram shows how objects interact with each other in a specific order. It focuses on the flow of messages or actions over time.

System Title

Laundry Pickup Scheduling System

Short Description

The Laundry Pickup Scheduling System is a simple application that lets customers book laundry pickup and delivery online. Customers can choose a pickup date, enter their location, and send a service request through the system. Staff members check the requests, confirm schedules, and update the status of each order. This system helps organize laundry services, avoids schedule conflicts, and makes it more convenient for customers.

Main Features

- Customer registration and login
- Schedule laundry pickup
- View booking status
- Staff approval of pickup requests
- Update delivery or completion status

Part 2: UML Diagram Creation

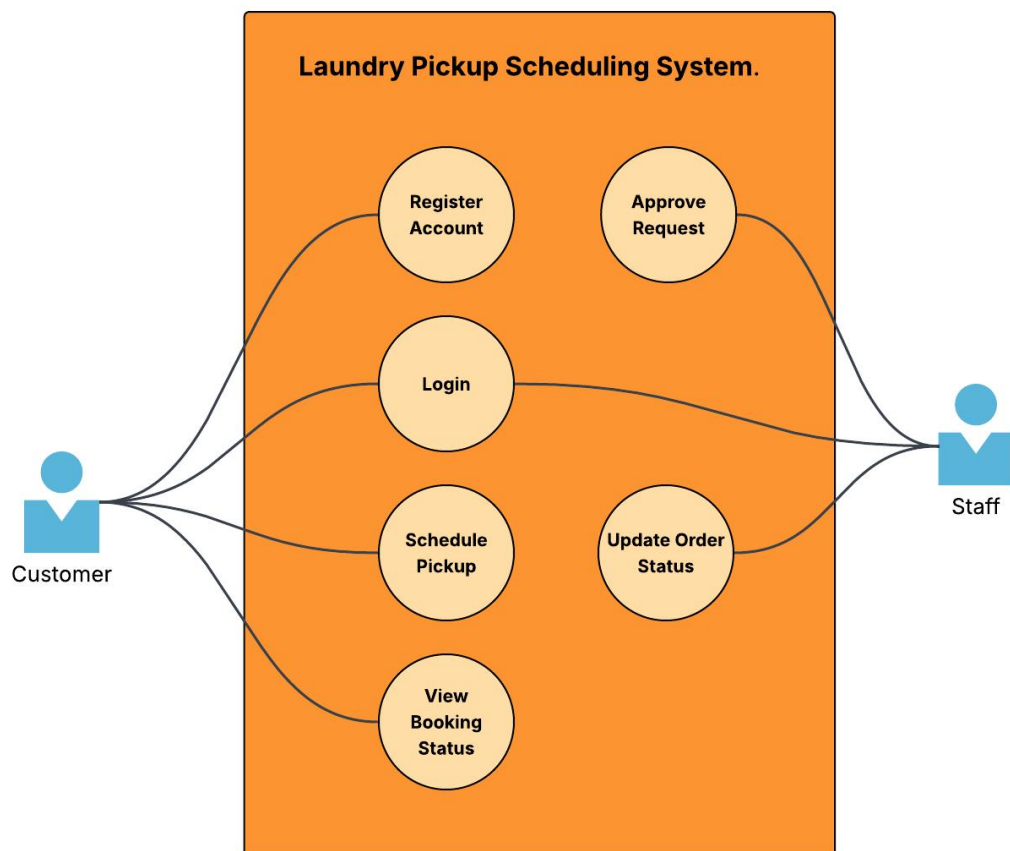
1. Use Case Diagram

Actors

- Customer
- Staff

Use Cases

- Register account
- Login
- Schedule pickup
- View booking status
- Approve request
- Update order status



Class Diagram

Customer

Attributes:

- customer_id
- name
- email
- password
- phone
- address

Methods:

- register()
- login()
- schedulePickup()
- viewBookingStatus()

Staff

Attributes:

- staff_id

- name
- email
- password
- role

Methods:

- login()
- approveRequest()
- updateOrderStatus()

Booking

Attributes:

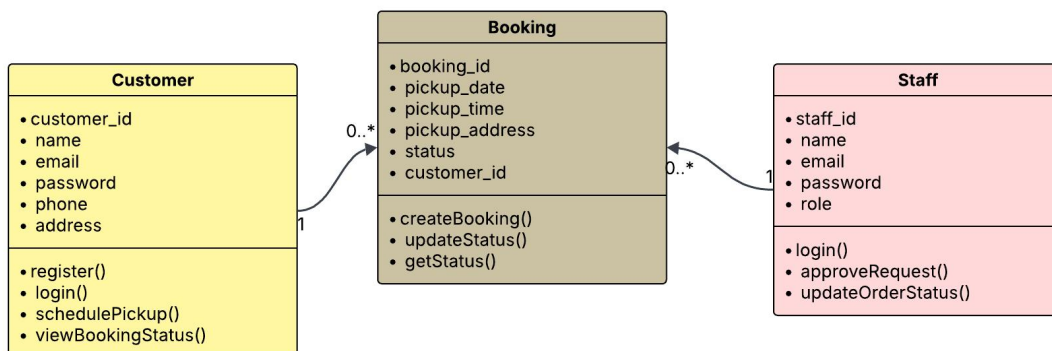
- booking_id
- pickup_date
- pickup_time
- location
- status
- customer_id

Methods:

- createBooking()
- updateStatus()
- getStatus()

UML Class Diagram

Laundry Pickup Scheduling System



Type something

Software Development Life Cycle (SDLC) Explanation

Planning

This is where the idea of the system is formed. The requirements are identified, and the purpose of the Laundry Pickup Scheduling System is defined.

Design

In this phase, the structure of the system is planned. UML diagrams are made to show how the system will work and how users interact with it.

Implementation

This is the actual development of the system. The features like booking a pickup, registration, and updating order status are created.

Testing

This phase is used to check if the system works properly. Errors and bugs are found and fixed before the system is finalized.

Part 4: Reflection

1. What did you learn from creating UML diagrams?

I learned how to properly show how a system works using diagrams. It helped me understand the flow of the system and how users interact with it.

2. Why is planning (SDLC) important before building a system?

Planning is important because it gives direction before starting the system. It helps avoid mistakes and makes the development process more organized.

Outputs:

Link for the Diagram:

1. UML Class Diagram - https://lucid.app/lucidchart/06fd99e2-9d22-449a-baa4-bba4805d8b84/edit?invitationId=inv_e1dced3b-dad2-468a-b5e4-170dbfb31787

2. Use Case Diagram - https://lucid.app/lucidchart/a42e1790-6c98-4242-a90e-420115d2a9fd/edit?viewport_loc=-1728%2C-817%2C3921%2C1737%2C0_0&invitationId=inv_85beade9-f771-4df4-b436-51945ae876b7

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Software Development Life Cycle (SDLC)

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